BANK CUSTOMER CHURN ANALYSIS

Business Problem & Retention Risk Analytics Brief

A decision-support project to identify churn patterns, prioritize retention investigation, and evaluate a baseline risk-classification model.

10,000	20.38%	3	18
Customer records	Historical churn rate	Geographies	Source columns

1. Business context

Customer churn reduces recurring customer value and can increase the cost of replacing lost relationships. A bank needs a disciplined way to understand which customer groups show elevated historical churn, what observable behaviours are associated with that outcome, and where retention teams should investigate first. A broad, untargeted campaign may spend effort on customers who are unlikely to leave while missing concentrated pockets of risk.

Business problem statement

The bank does not have a clear, evidence-led view of which customer profiles and engagement patterns are associated with account exit. It needs to convert historical customer data into actionable retention segments and assess whether a baseline model can rank customers by observed churn risk.

2. Decision the analysis should support

• Who should be investigated first? Identify customer segments with higher historical churn rates while showing the number of customers in each segment.

• What should the business examine? Highlight potential issues around product holding, engagement, account balances, age bands, and regional differences.

• Can risk be ranked consistently? Compare classification models using churn-focused metrics and make the trade-off between missed churners and false alerts visible.

3. Dataset scope

Item	Scope / definition
Source	Kaggle - Bank Customer Churn dataset supplied for this project.
Population	10,000 customer records across France, Germany, and Spain.
Outcome	`Exited`: 1 = churned / exited; 0 = retained. Prior notebook run: 2,038 exited and 7,962 retained.
Predictor domains	Credit score, geography, gender, age, tenure, balance, product count, card ownership, active membership, estimated salary, card type, and points earned.
Excluded / audited fields	Row identifiers and surname are not predictors. Complaint status is leakage-prone. Satisfaction score is descriptive-only until its timing is confirmed.

4. Initial evidence from the existing analysis

The figures below are taken from the previously executed notebook and supplied dashboard/README. They are baseline findings to be revalidated after running the refreshed workflow; they are not represented as freshly recomputed results.

Dimension	Observed result	Business interpretation
Geography	Germany 32.44%; Spain 16.67%; France 16.17%.	Germany is the clearest regional outlier in this sample and merits a focused diagnostic review.
Age	Mean age: churned 44.8 vs retained 37.4.	Older customers show higher observed churn; examine service needs and product fit rather than assuming a cause.
Product count	1 product 27.71% (n=5,084); 2 products 7.60% (n=4,590); 3 products 82.71% (n=266); 4 products 100% (n=60).	The pattern is non-linear. The 3- and 4-product groups are small; investigate the journey and validate stability before broad policy changes.
Engagement	Inactive 26.87%; active 14.27%.	Inactivity is a practical segmentation signal for a re-engagement test.
Balance + activity	Inactive + balance 31.66%; active + zero balance 9.61%.	A combined segment may be more useful for prioritizing outreach than a single attribute.
Satisfaction	Churn rates 19.64%-21.80%; chi-square p=0.433.	No significant association was detected in the prior analysis; do not overstate survey score as a churn driver.

5. Predictive model baseline

The prior notebook used a stratified 80/20 train-test split. Complaint status was excluded because of its near-identity with the target. The table below reproduces the previous run; the refreshed pipeline adds preprocessing controls and Average Precision (PR-AUC), so its results may differ after execution.

Metric	Logistic Regression	Random Forest
Accuracy	0.81	0.82
Precision - churn	0.62	0.55
Recall - churn	0.21	0.72
ROC-AUC	0.78	0.87

Interpretation: Random Forest found 72% of churners in the holdout sample, compared with 21% for Logistic Regression, at the default threshold. Its churn precision was lower (55%), so it also generated false alerts. The threshold must be chosen with retention-team capacity and the relative cost of missed churners in mind.

6. Data quality and interpretation controls

• Leakage: 2,034 of 2,044 customers with `Complain = 1` exited (99.51%). Complaint status is a late-stage/target-proxy field in this sample and must not be used as a predictor.

• Timing uncertainty: the dataset does not provide verified feature timestamps or a defined lead time before exit. It cannot support a specific 30-, 60-, or 90-day churn forecast claim.

• Sample effects: extreme rates in small groups, especially four products (n=60), should be displayed with counts and validated on new data.

• Association vs causation: regional, age, balance, and product patterns identify where to investigate; they do not prove why customers left.

Scope and provenance

Portfolio analysis using the public Kaggle Bank Customer Churn dataset. Baseline figures came from the supplied notebook, README and two-page dashboard; revalidate them by rerunning the refreshed notebook with the CSV attached. The refreshed workflow has not yet been executed against the source CSV.

7. Recommended business actions

01 | Review the multi-product journey

Audit onboarding, bundling, servicing and customer communications around customers with three or more products. Start with a small diagnostic sample; do not change product strategy solely on the 60-person four-product segment.

02 | Test a focused Germany retention review

Segment German customers by balance, tenure, product count and engagement. Validate whether the elevated churn persists after controlling for customer mix. Explore relationship-manager feedback and competitor context as hypotheses, not conclusions.

03 | Pilot inactivity-based outreach

Design a controlled re-engagement test for inactive customers, with a separate view of customers who still hold a balance. Measure incremental retention against a comparable control group.

04 | Reassess the satisfaction measurement

Check survey timing, response coverage and whether the score records complaint resolution rather than general relationship satisfaction. Consider adding behavioural and service-recovery measures.

05 | Use model scores as a prioritization aid only

Compare thresholds using precision, recall, false alerts and missed churners. Before operational use, validate on a later time period or a fresh cohort, review calibration and fairness, and obtain business approval for the outreach policy.

8. How success should be measured

Measure	Recommended evaluation
Retention impact	Incremental churn reduction in a controlled campaign vs. a comparable control group.
Targeting efficiency	Precision among contacted customers; churners identified per 100 outreach attempts.
Coverage	Recall of churners at the selected outreach capacity / threshold.
Commercial value	Retained customer value less incentives, contact cost, and servicing cost. Requires bank-supplied financial data.
Model governance	Performance by geography and key customer segments; drift, calibration, fairness and data-quality checks.

9. Analytical workflow and deliverables

Workflow: data audit and conservative cleaning -> exploratory churn analysis -> statistical validation -> SQL cross-checks -> leakage-aware model comparison -> threshold analysis -> Power BI refresh -> business interpretation and recommendations.

Planned artefacts: cleaned customer dataset; churn breakdown tables with denominators; SQL validation outputs; model comparison metrics; threshold sensitivity table; holdout-only churn probability scores; interactive Power BI report; documented limitations and recommendations.